

❖ End-to-End ML Pipeline with Ethical Auditing and Generative AI

Breast Cancer Prediction Capstone

GitHub Link: https://github.com/DreamerZ-2025/capstone_healthcare_ai





□ Problem Framing:

- ❖ False negatives cost lives and increase late-stage treatment costs.
- ❖ ML Problem: Binary Classification (Malignant vs. Benign).
- ❖ Success Metrics: Maximize Recall ($> 95\%$) and ROC-AUC.



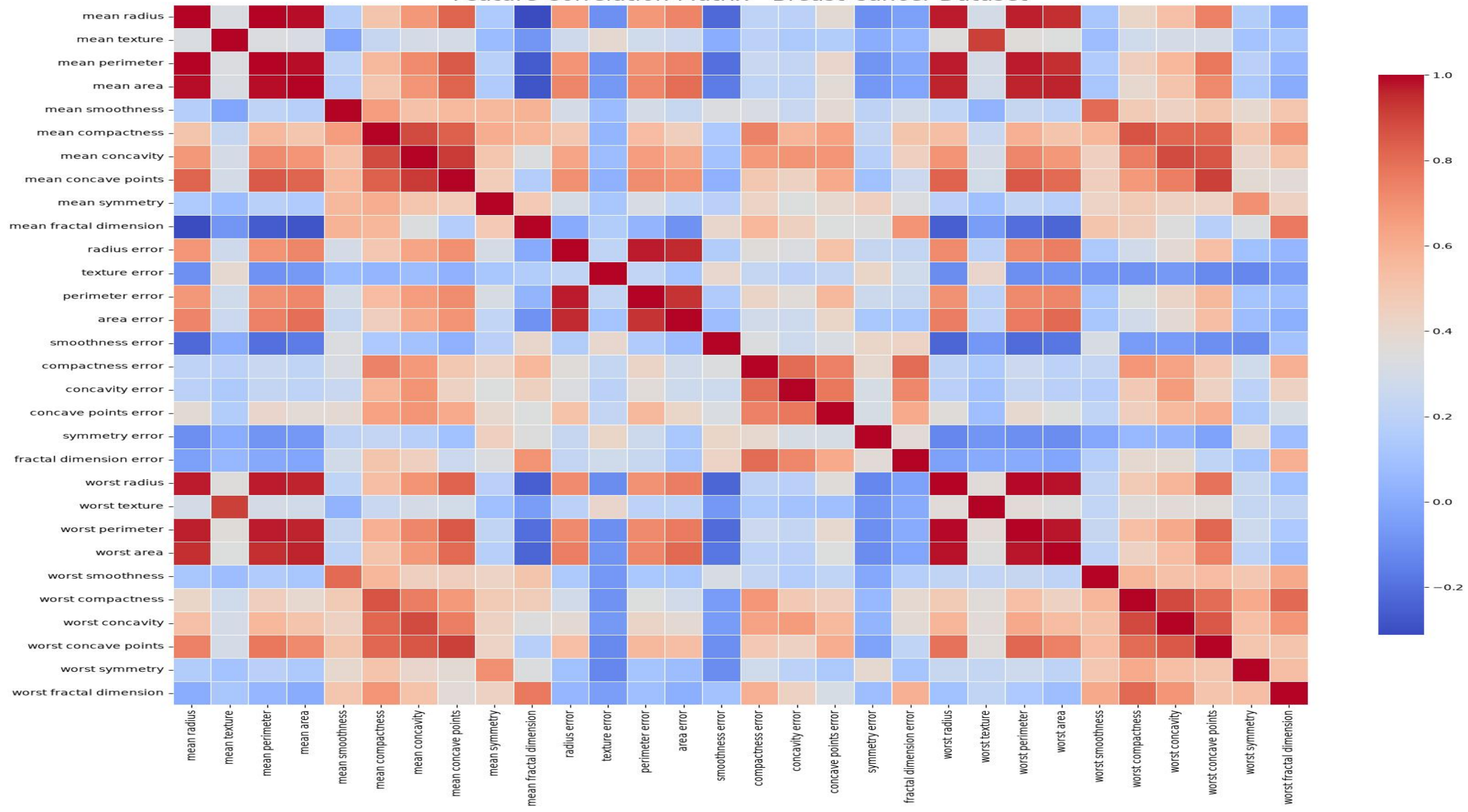


□ Dataset Overview & EDA:

- ❖ **Source:** `sklearn.datasets.load_breast_cancer`
- ❖ **569 samples, 30 numerical features.**
- ❖ **Class Distribution: 37% Malignant, 63% Benign.**



Feature Correlation Matrix - Breast Cancer Dataset





□ Data Preprocessing:

- ❖ **Stratified Train/Test Split (80/20) to maintain class balance.**
- ❖ **StandardScaler applied to normalize feature ranges.**
- ❖ **Why: Essential for distance-based algorithms and gradient descent convergence.**





□ Model Architecture & Tuning:

- ❖ **Models Evaluated: Logistic Regression (Baseline) vs. Random Forest.**
- ❖ **Hyperparameter Tuning: GridSearchCV with 5-fold Stratified Cross-Validation.**
- ❖ **Best Params (LR): C=1. Best Params (RF): n_estimators=200, max_depth=None.**



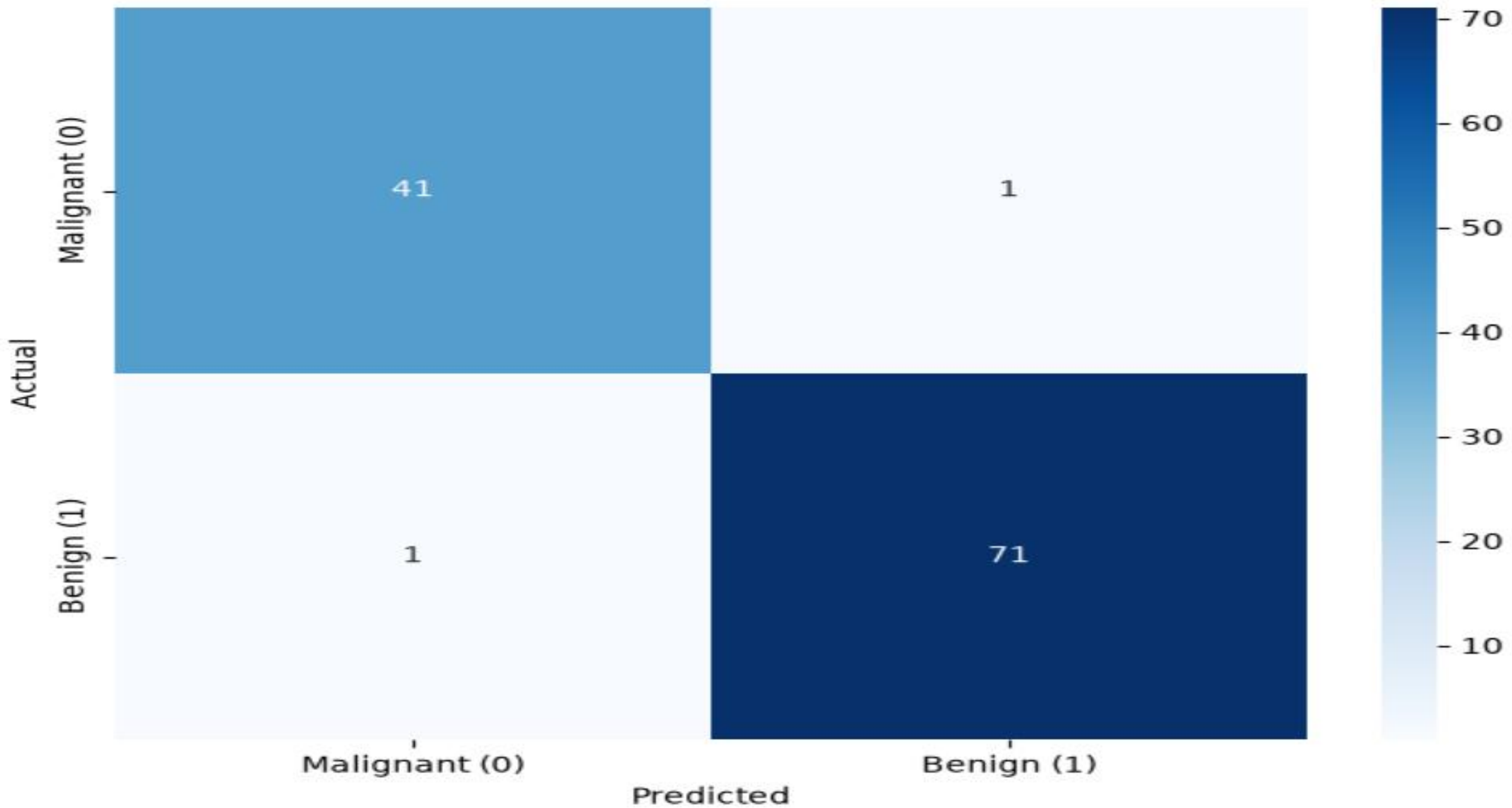


❑ Model Evaluation Results:

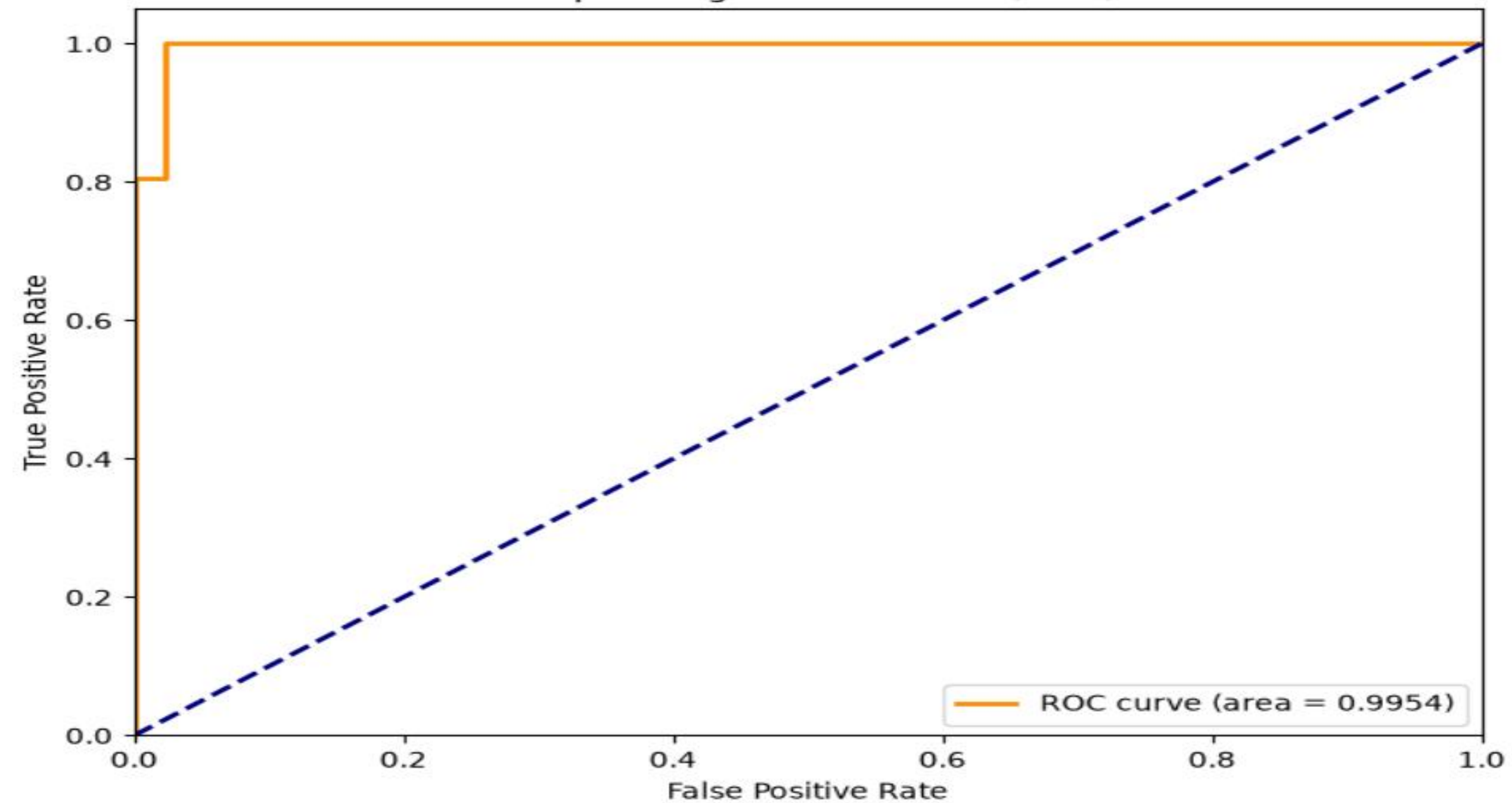
- ❖ **Winner: Logistic Regression (ROC-AUC: 0.9954).**
- ❖ **Recall (Malignant): 0.98**
- ❖ **F1-Score (Malignant): 0.96**



Confusion Matrix - Breast Cancer Prediction



Receiver Operating Characteristic (ROC) Curve



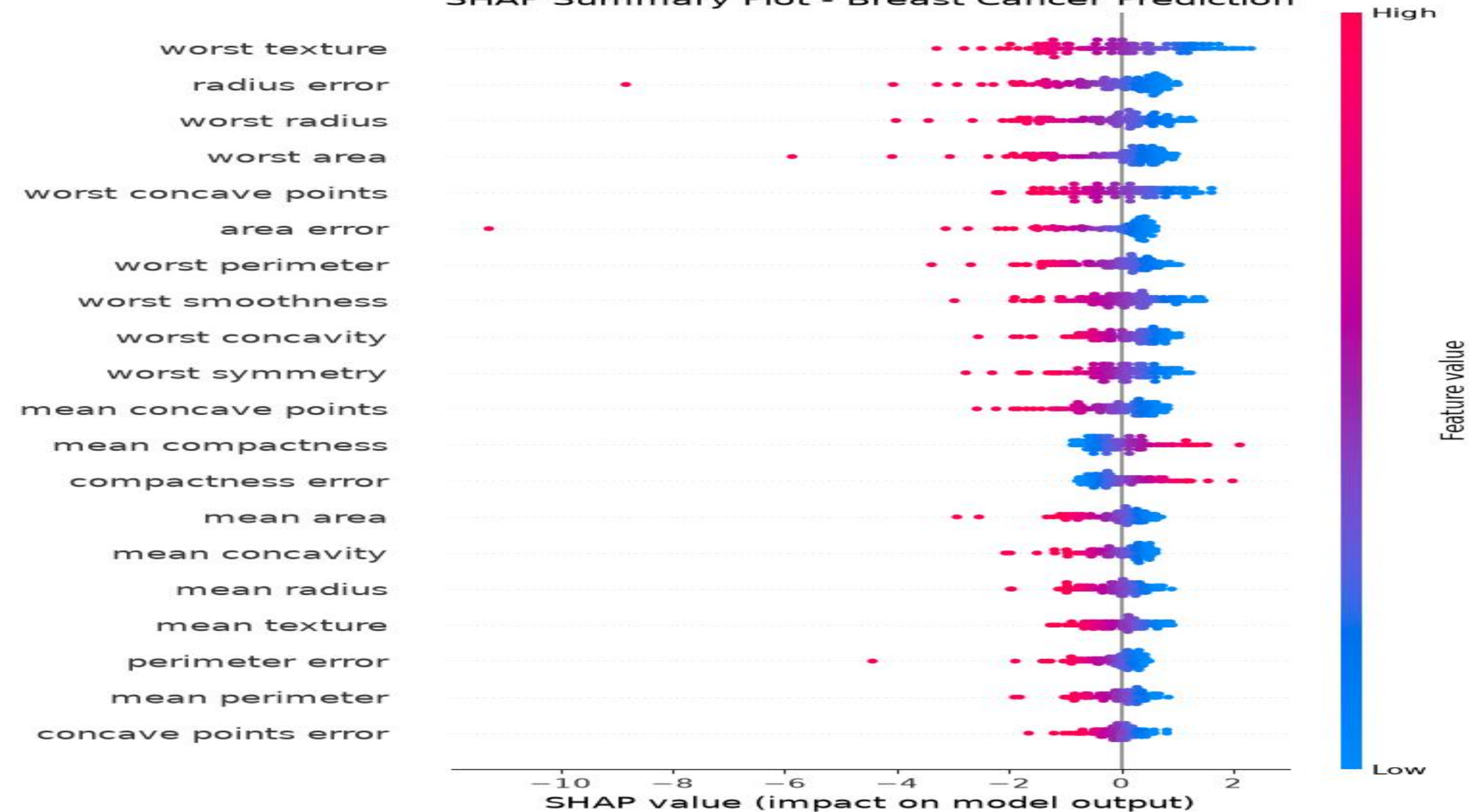


□ Explainability (SHAP):

- ❖ Used SHAP (SHapley Additive exPlanations) to interpret model decisions.
- ❖ Key Finding: 'Worst Area', 'Worst Perimeter', and 'Mean Concave Points' are the strongest predictors of malignancy.



SHAP Summary Plot - Breast Cancer Prediction





□ Ethical AI & Bias Audit:

- ❖ Proxy Attribute: Tumor Size (Large vs. Small).
- ❖ Disparate Impact Ratio (DIR): 1.09.
- ❖ Conclusion: No significant disparate impact detected (DIR is within 0.80 - 1.25 range).





❑ Deployment Architecture:

- ❖ **Backend: FastAPI REST API (/predict and /ask endpoints).**
- ❖ **Frontend: PySide6 Desktop GUI client.**
- ❖ **Generative AI: RAG pipeline using LangChain, FAISS, and Google Gemini.**





□ Conclusion & Future Work:

- ❖ Successfully built a high-recall, ethically-audited ML model.
- ❖ Deployed with a robust API and user-friendly GUI.
- ❖ Future Work: Validate on real-world demographic data, integrate with hospital EMR systems.

